

Assignment 6

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For this assignment, I used the fashion-mnist dataset which is very similar to the fashion dataset. It has 28x28 pixel images, 10 outputs which are:

- 0 T-shirt/top
- 1 Trouser
- 2 Pullover
- 3 Dress
- 4 Coat
- 5 Sandal
- 6 Shirt
- 7 Sneaker
- 8 Bag
- 9 Ankle boot

The data also consists of 70,000 rows.

I split the code into 60,000 for training, 10,000 for validation, and 5,00 for testing. Then I ran the initial mean squared error for the data without training the model. This number wasn't completely aligned with the 10% mark for the 10 outputs, but it was similar enough so I assumed no issue there.

```
Initial valid MSE: 0.3  
Initial valid accuracy: 7.7%
```

Then to test the mean squared error function, I ran lines 158-162. I ended up with the same numbers once again meaning my function was working correctly.

Then, I was onto the actual training of the data. I began with the default settings- a batch size of 100, learning rate of 0.1, epoch of 20 and hidden layer of 50. The results were pretty good, however before even looking at any charts, the scores were not horrible but not great, and I could tell that there was a large discrepancy between the training and validation set.

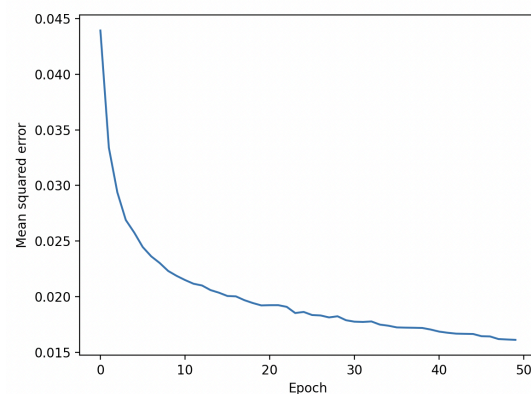
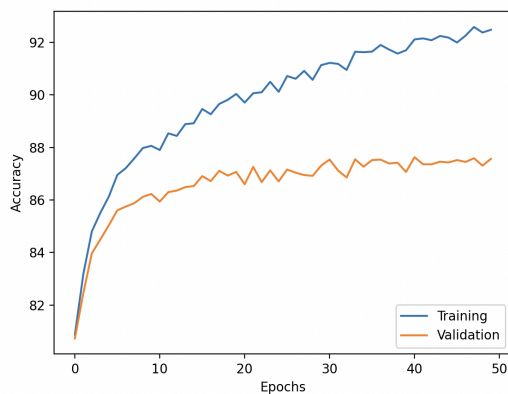
Epoch: 030/050	Train MSE: 0.02	Train Acc: 87.43%	Valid Acc: 85.92%
Epoch: 031/050	Train MSE: 0.02	Train Acc: 87.48%	Valid Acc: 85.95%
Epoch: 032/050	Train MSE: 0.02	Train Acc: 87.40%	Valid Acc: 85.93%
Epoch: 033/050	Train MSE: 0.02	Train Acc: 87.46%	Valid Acc: 85.93%
Epoch: 034/050	Train MSE: 0.02	Train Acc: 87.70%	Valid Acc: 86.09%
Epoch: 035/050	Train MSE: 0.02	Train Acc: 87.76%	Valid Acc: 86.08%
Epoch: 036/050	Train MSE: 0.02	Train Acc: 87.77%	Valid Acc: 86.27%
Epoch: 037/050	Train MSE: 0.02	Train Acc: 87.84%	Valid Acc: 86.28%
Epoch: 038/050	Train MSE: 0.02	Train Acc: 87.92%	Valid Acc: 86.34%
Epoch: 039/050	Train MSE: 0.02	Train Acc: 88.01%	Valid Acc: 86.11%
Epoch: 040/050	Train MSE: 0.02	Train Acc: 88.04%	Valid Acc: 86.34%
Epoch: 041/050	Train MSE: 0.02	Train Acc: 88.16%	Valid Acc: 86.38%
Epoch: 042/050	Train MSE: 0.02	Train Acc: 88.18%	Valid Acc: 86.43%
Epoch: 043/050	Train MSE: 0.02	Train Acc: 88.28%	Valid Acc: 86.55%
Epoch: 044/050	Train MSE: 0.02	Train Acc: 88.33%	Valid Acc: 86.57%
Epoch: 045/050	Train MSE: 0.02	Train Acc: 88.37%	Valid Acc: 86.42%
Epoch: 046/050	Train MSE: 0.02	Train Acc: 88.36%	Valid Acc: 86.53%
Epoch: 047/050	Train MSE: 0.02	Train Acc: 88.38%	Valid Acc: 86.50%
Epoch: 048/050	Train MSE: 0.02	Train Acc: 88.57%	Valid Acc: 86.58%
Epoch: 049/050	Train MSE: 0.02	Train Acc: 88.64%	Valid Acc: 86.63%
Epoch: 050/050	Train MSE: 0.02	Train Acc: 88.66%	Valid Acc: 86.60%

I decided then to change the learning rate to 0.5, since my dataset was a little larger than the mnist. This improved the scores by a lot however there was an even greater discrepancy as represented in the chart. The shape of the loss function also did not change much.

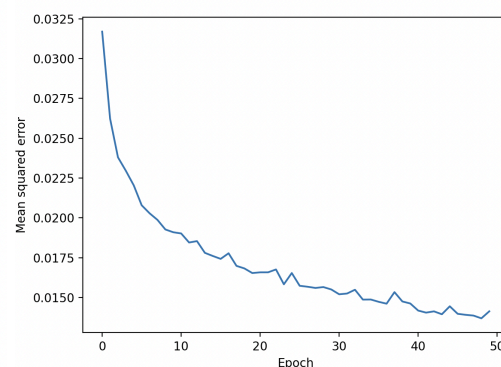
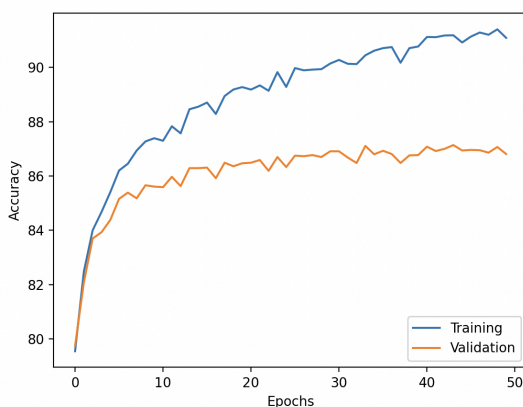
```

Epoch: 041/050 | Train MSE: 0.01 | Train Acc: 92.11% | Valid Acc: 87.63%
Epoch: 042/050 | Train MSE: 0.01 | Train Acc: 92.15% | Valid Acc: 87.36%
Epoch: 043/050 | Train MSE: 0.01 | Train Acc: 92.08% | Valid Acc: 87.36%
Epoch: 044/050 | Train MSE: 0.01 | Train Acc: 92.24% | Valid Acc: 87.46%
Epoch: 045/050 | Train MSE: 0.01 | Train Acc: 92.18% | Valid Acc: 87.43%
Epoch: 046/050 | Train MSE: 0.01 | Train Acc: 92.00% | Valid Acc: 87.52%
Epoch: 047/050 | Train MSE: 0.01 | Train Acc: 92.26% | Valid Acc: 87.45%
Epoch: 048/050 | Train MSE: 0.01 | Train Acc: 92.59% | Valid Acc: 87.59%
Epoch: 049/050 | Train MSE: 0.01 | Train Acc: 92.38% | Valid Acc: 87.31%
Epoch: 050/050 | Train MSE: 0.01 | Train Acc: 92.48% | Valid Acc: 87.57%

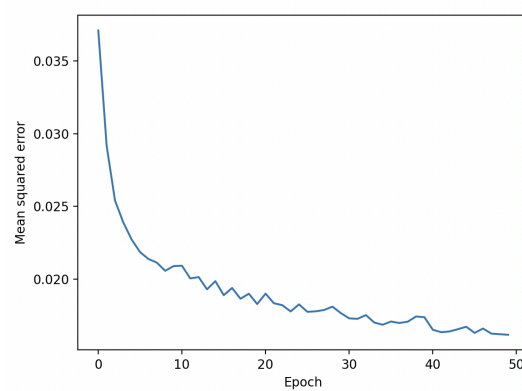
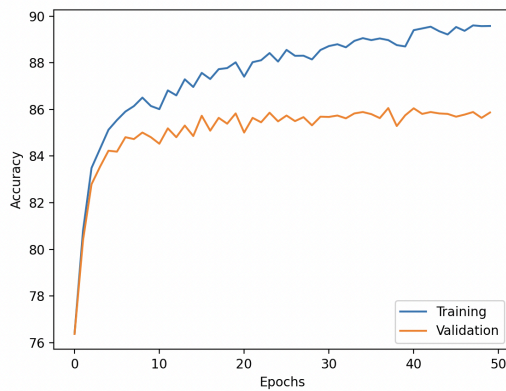
```



To reduce the overfitting rate, I tried lowering the hidden layers to 30 instead of 20. This did not have much of an impact on the model. Overfitting slightly decreased but not by much. There was also a slightly less smooth convergence which is likely because the model had less steps to take.

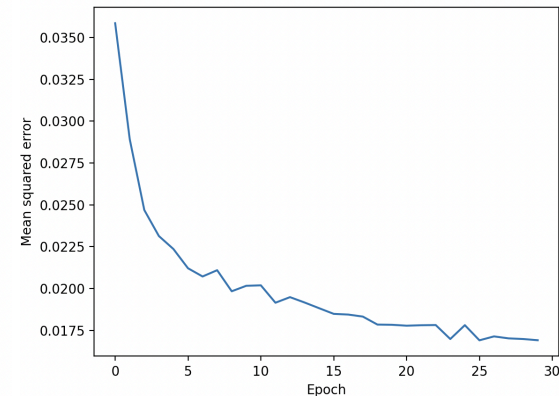
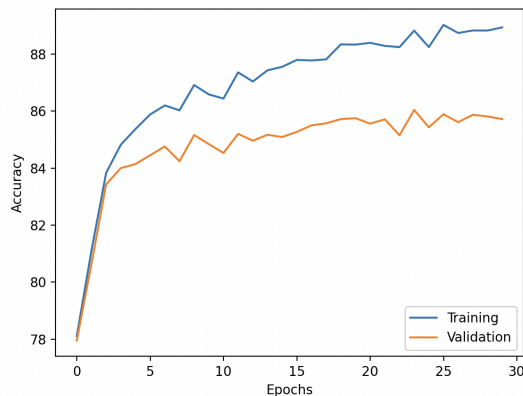


I tried lowering the number of hidden layers again to see if there would be any change, this time to 15. It overfit slightly less and the model was converging quicker, however the overall score decreased this time to 89% and 85%.



```
Epoch: 044/050 | Train MSE: 0.02 | Train Acc: 89.36% | Valid Acc: 85.83%
Epoch: 045/050 | Train MSE: 0.02 | Train Acc: 89.22% | Valid Acc: 85.81%
Epoch: 046/050 | Train MSE: 0.02 | Train Acc: 89.54% | Valid Acc: 85.69%
Epoch: 047/050 | Train MSE: 0.02 | Train Acc: 89.37% | Valid Acc: 85.78%
Epoch: 048/050 | Train MSE: 0.02 | Train Acc: 89.61% | Valid Acc: 85.89%
Epoch: 049/050 | Train MSE: 0.02 | Train Acc: 89.58% | Valid Acc: 85.64%
Epoch: 050/050 | Train MSE: 0.02 | Train Acc: 89.58% | Valid Acc: 85.87%
```

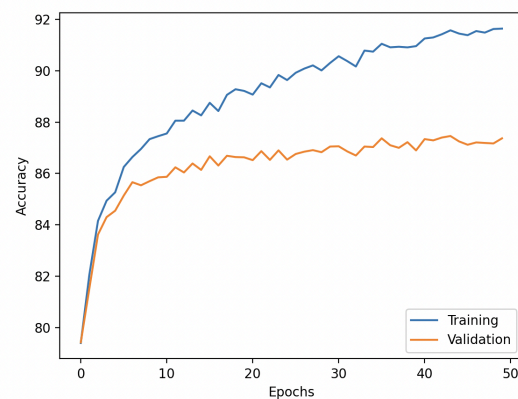
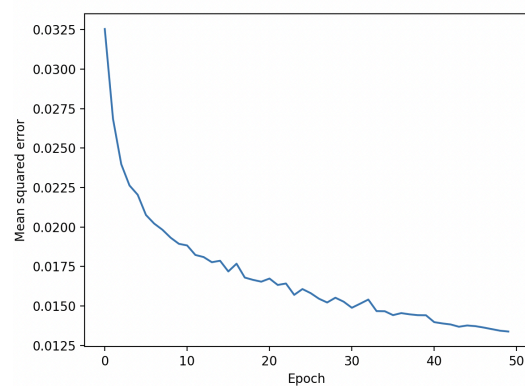
To get the model to overfit less, I decided to play around with some hyperparameter values, starting with epochs. The idea was that the less iterations I have the less room the model has space to overfit. I increased the hidden layers to 20 and changed the epoch to 30. This was the worst of the bunch.



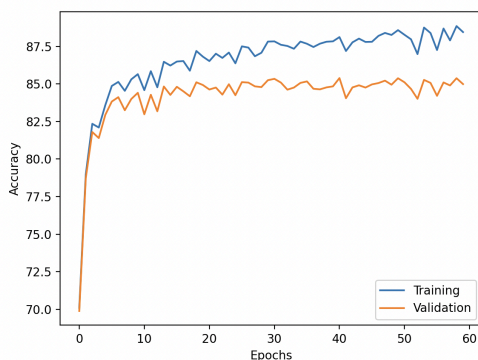
```
Epoch: 026/030 | Train MSE: 0.02 | Train Acc: 89.02% | Valid Acc: 85.89%
Epoch: 027/030 | Train MSE: 0.02 | Train Acc: 88.74% | Valid Acc: 85.61%
Epoch: 028/030 | Train MSE: 0.02 | Train Acc: 88.83% | Valid Acc: 85.87%
Epoch: 029/030 | Train MSE: 0.02 | Train Acc: 88.82% | Valid Acc: 85.81%
Epoch: 030/030 | Train MSE: 0.02 | Train Acc: 88.94% | Valid Acc: 85.72%
```

For the final iteration I decided to increase the epoch to 50 and lower the hidden layers back up to 40, and a learning rate of learning rate of 0.3. This was slightly better, but the model was still heavily overfitting. This time more than the last two attempts, however the score was higher.

```
Epoch: 044/050 | Train MSE: 0.01 | Train Acc: 91.58% | Valid Acc: 87.46%
Epoch: 045/050 | Train MSE: 0.01 | Train Acc: 91.45% | Valid Acc: 87.25%
Epoch: 046/050 | Train MSE: 0.01 | Train Acc: 91.39% | Valid Acc: 87.12%
Epoch: 047/050 | Train MSE: 0.01 | Train Acc: 91.55% | Valid Acc: 87.21%
Epoch: 048/050 | Train MSE: 0.01 | Train Acc: 91.49% | Valid Acc: 87.19%
Epoch: 049/050 | Train MSE: 0.01 | Train Acc: 91.63% | Valid Acc: 87.17%
Epoch: 050/050 | Train MSE: 0.01 | Train Acc: 91.64% | Valid Acc: 87.37%
```



For the last attempt, I decided to decrease the hidden layers all the way back down to 10, learning rate up to 0.5, but increase the epochs to 60.



Even though the model was not as good as it could be, I decided to run the tests on the model. The final accuracy was 85.19%, this suggests that it overfitted even further between the validation and testing model. I also ran the visualization that detailed the specific accuracy for each type of clothing. It seems as if it had the hardest time predicting short sleeve shirts.



Test accuracy: 85.19%